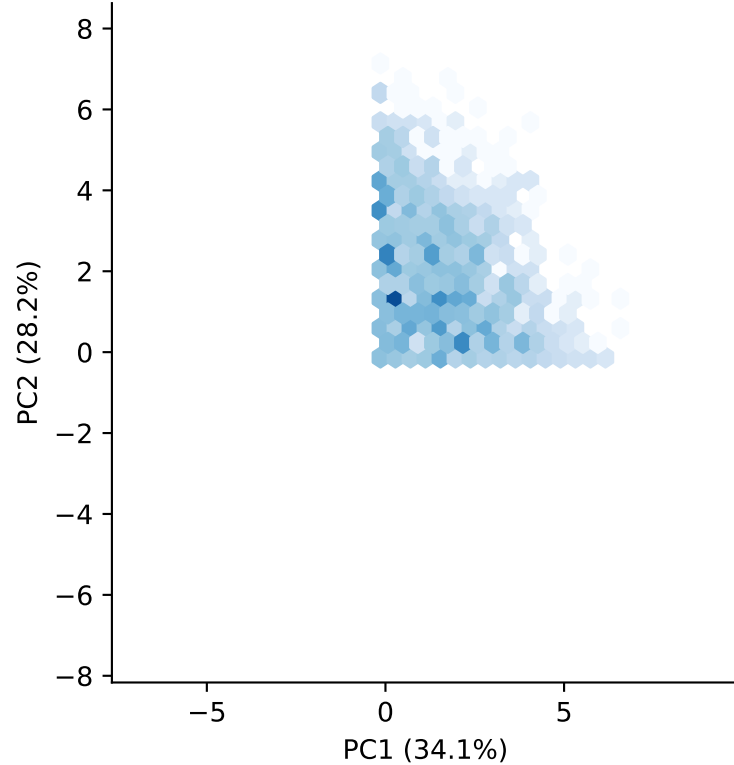
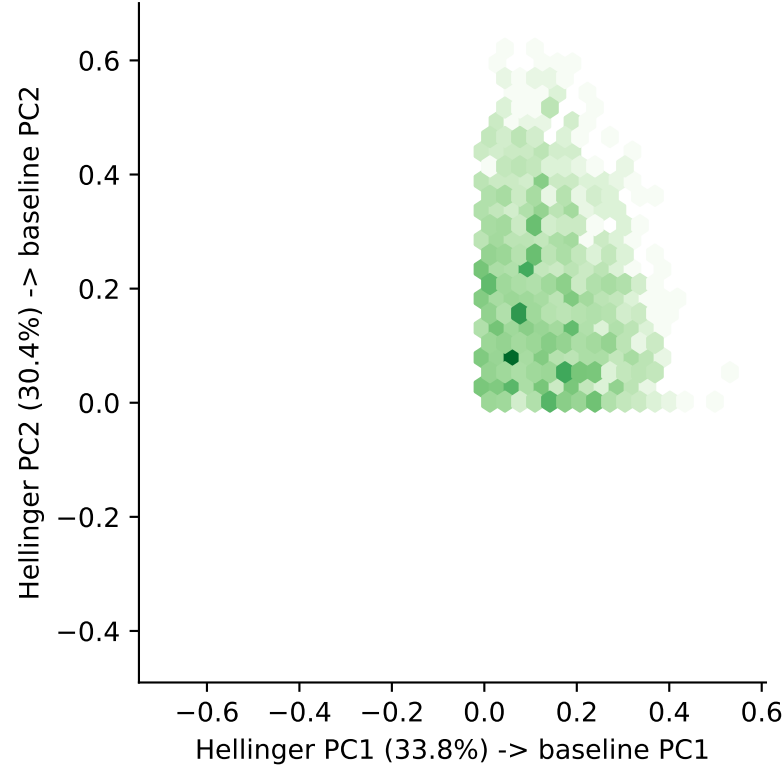


Fire VASE compositional-geometry sensitivity

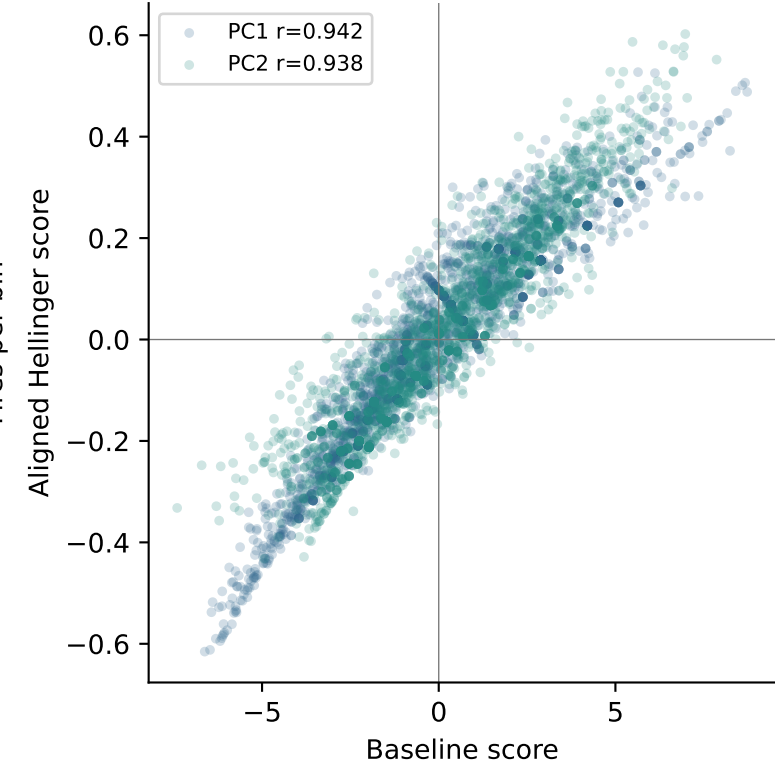
A Baseline standardized-Euclidean PCA



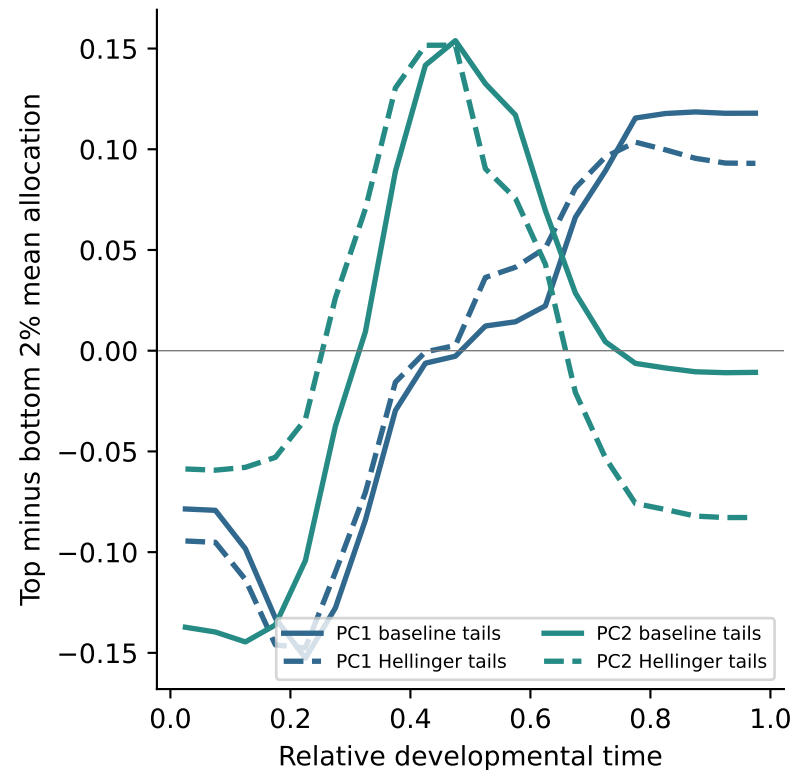
B Hellinger PCA, aligned



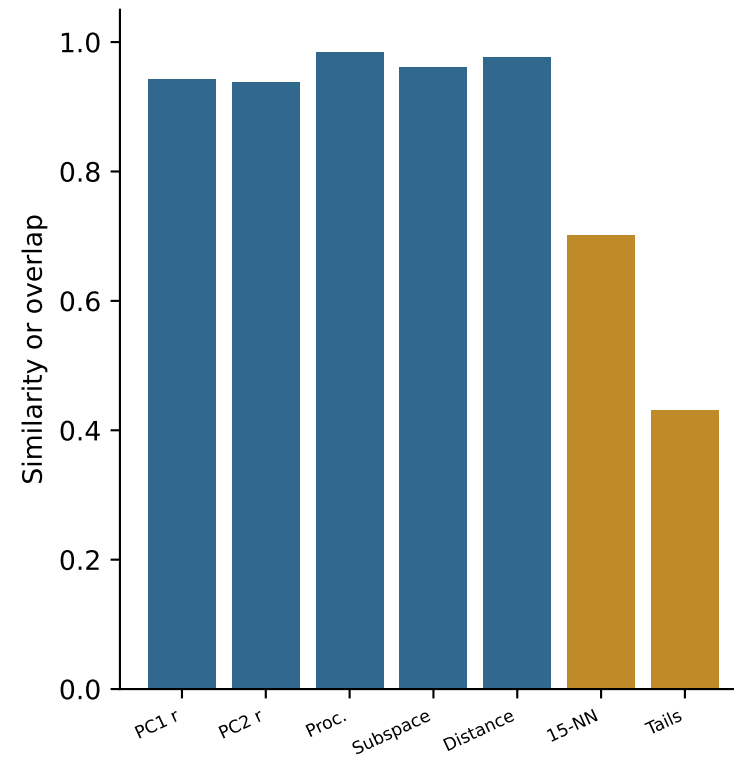
C Aligned score relationships



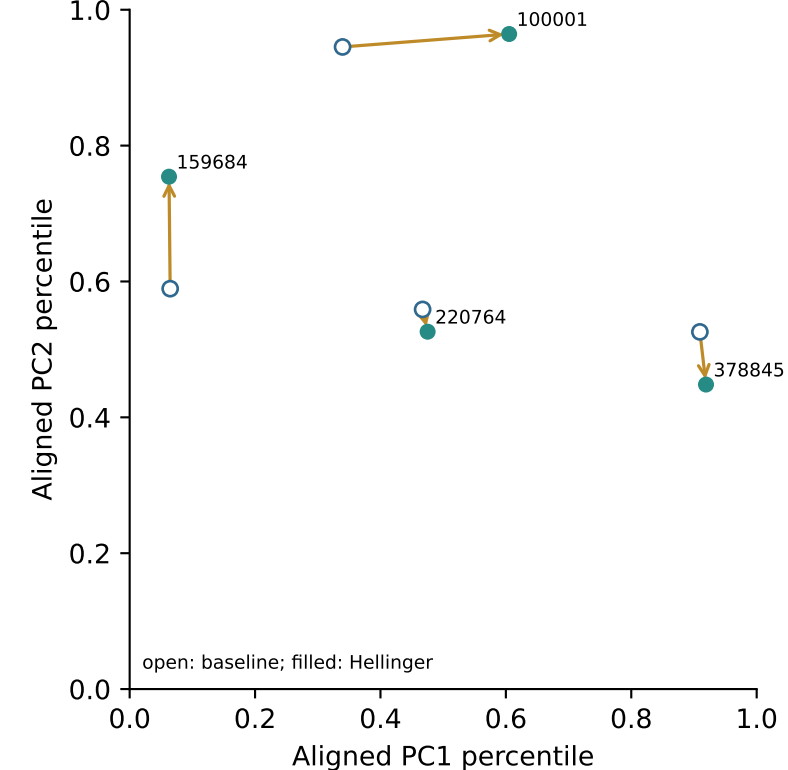
D Raw-allocation tail contrasts



E Global and local stability



F Figure 1/2 exemplar movement



Same 10,246 gap-free fires and 20 allocation bins. Hellinger PCA uses $\sqrt{\text{allocation}}$, mean centering, and no variance scaling. Loading magnitudes are not compared across units.